edunet foundation

Getting Started with Cybersecurity

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DISCLAIMER

Learning Objective

- Cybersecurity Fundamentals and CIA Triad
- Sensitive Data Protection
- Cyber Threats and Attacks
- Security Tools and Malware Protection
- VPNs and Online Privacy
- Cybersecurity Careers and Certifications
- Skills and Job Market Demand in Cybersecurity

Why This Matters Now

32%

Projected growth in cybersecurity jobs by 2032 [1]

4 million

Cybersecurity positions unfilled globally [1]

3.4 million

Cybersecurity jobs currently remain unfilled [2]

35%

Expected growth rate by 2031 [2]

*The **demand** far exceeds **supply** , creating unprecedented **career opportunities** for new professionals*

Part 1: Cybersecurity Fundamentals

Understanding Cybersecurity Threats

Threats

Anything leading to harmful events affecting systems or data

Vulnerabilities

Weaknesses in hardware, firmware, or software that hackers exploit

Attackers

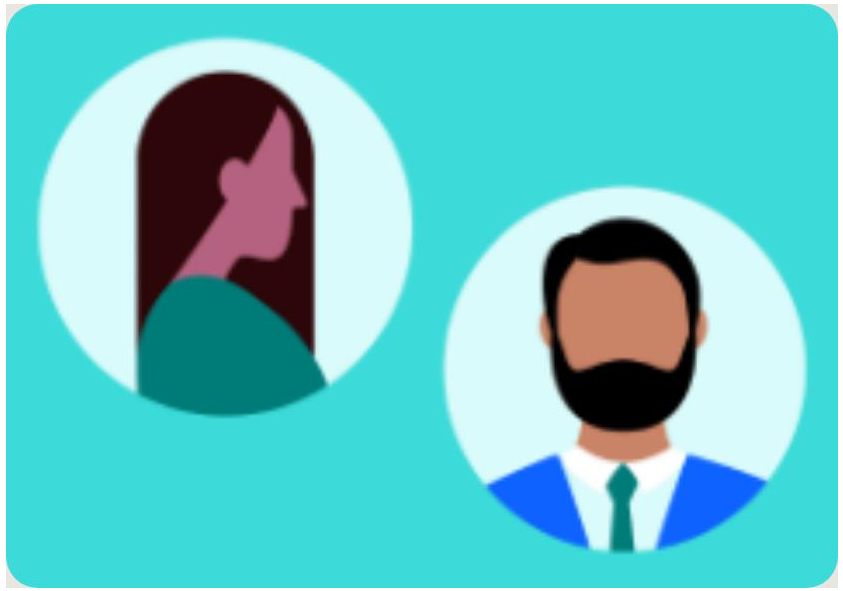
Hackers bypassing security for malicious purposes like data theft

Malware

Malicious software including viruses, worms, ransomware, and trojans

- A threat is anything that could lead to harmful events affecting your network, system, or data [2]
- Vulnerabilities include outdated devices without security updates, creating entry points for attackers to exploit
- Attackers use social engineering tactics like fake emails from trusted sources to trick users into revealing passwords
- Trojans disguise themselves as helpful programs while secretly giving attackers remote access to your device and data

The Three Pillars of Cybersecurity



Effective cybersecurity requires people, processes, and technology working together

- People are the most essential part as they design, maintain, and monitor systems while also being potential weak links
- Processes must be clear, accessible, consistent, and easy to follow without creating frustration for users
- Technology includes infrastructure and protective tools like antimalware, firewalls, and email security when implemented correctly
- Cybersecurity protection involves multiple layers combining all three elements in a coordinated defense strategy [2]

People: The Human Element

Humans are both your strongest defense and your greatest vulnerability

- People are responsible for using technology securely, designing and maintaining systems, and monitoring security alerts effectively [2]
- All employees must understand why cybersecurity matters and their specific role in protecting organizational systems and information [2]
- Training should be appropriate for each employee's job role and industry to address relevant threats and scenarios
- A good cybersecurity approach anticipates human behavior and designs systems that accommodate it rather than fighting against it [2]

Processes and Technology

Processes:

- Security integrated at each step with clear, accessible, consistent procedures
- Good processes avoid complexity, contradictions, and excessive exceptions that frustrate users

Technology:

- Infrastructure tools include antimalware, email security, firewalls, and device management
- Properly implemented technology remains non-intrusive and barely noticeable [2]

Real-World Application: Home Networks

People

Knowledge of security measures, safe browsing, phishing detection

Process

Password complexity, router setup instructions, admin password changes

Technology

Password requirements, VPN encryption, connected devices, firewalls

- Consider who sets up network and their knowledge of strong passwords and security measures [2]
- Evaluate household members' safe browsing habits, phishing detection ability, and software updates [2]
- Router instructions should be clear and suggest changing default admin passwords with strengthening tips [2]
- Each connected device multiplies potential targets requiring adequate security on all endpoints [2]

Part 2: Data and Privacy

Understanding Data

Text, Numbers, Images

Data formats from digital sources

Government, Healthcare, Financial

Major data collectors

- Data includes text, numbers, measurements, statistics, facts, and images from computers, smartphones, networks, and online platforms [2]
- Government stores identifying data like Social Security numbers, driver's licenses, birthdates, addresses, and physical characteristics
- Healthcare providers maintain medical history, test results, diagnoses, and prescribed medications alongside identifying information
- Financial institutions track all transactions including date, time, location, company, amount, plus account credentials and personal details

Data Security vs Data Privacy

- Data security protects confidential data from unauthorized access, disclosure, or destruction including proprietary and personal information [2]
- Security encompasses all company data such as trade secrets, financial records, strategic plans, and employee information
- Data privacy is a specific type of data security focused only on customers' and employees' private personal data [2]
- Privacy addresses collection, disclosure, and use of personal information like addresses, Social Security numbers, and credit cards

Consequences of Data Breaches

- Organizations face penalties, fines, fees, legal settlements, and judgments that can total millions of dollars
- Companies must investigate breaches, upgrade security systems, revise policies, and dedicate extensive time to recovery
- Bad press coverage and public criticism damage brand trust and customer confidence for years following incidents
- The cost of prevention is always far less than the cost of breach recovery

Financial

Penalties, fines, fees, legal settlements totaling millions

Operational

Investigation, security upgrades, policy revisions, extensive recovery time

Reputational

Bad press coverage, public criticism, damaged brand trust for years

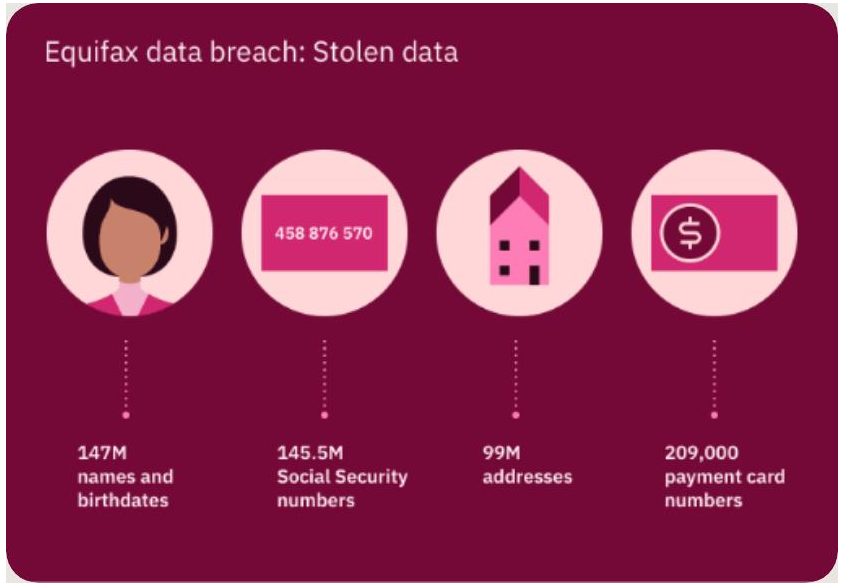
Case Study: Equifax Breach

$425M
Settlement paid to affected consumers [2]

147M
Names and birthdates compromised [2]

145.5M
Social Security numbers stolen [2]

- Attackers exploited unpatched vulnerability to access network with unencrypted credentials
- Breach also compromised 209,000 payment cards creating widespread financial fraud risk [2]
- Delayed disclosure and flawed consumer website rollout drew intense media criticism [2]
- Company spent $1.4M on security upgrades after the breach [2]



Framework

AICPA and CICA developed GAPP standard

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Principles

10 comprehensive best practices covering privacy

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Application

Organizations design policies, conduct audits, assess risks

Privacy Best Practices: GAPP

- The American Institute of CPAs and Canadian Institute of Chartered Accountants developed GAPP as a widely recognized privacy standard [2]
- The framework includes 10 comprehensive best practices covering all aspects of organizational data privacy management
- Organizations use GAPP principles to design privacy policies, conduct audits, assess risks, and investigate privacy complaints
- Following established frameworks helps organizations demonstrate compliance with privacy laws and regulations across jurisdictions

Part 3: Understanding Malware Threats

What is Malware?

Malware is designed to operate silently while exploiting your devices

- Attackers design malware to remain undetected while using device resources to launch attacks, host illicit data, or access information [1]
- The typical reason is financial gain, such as stealing personal financial information or borrowing computing power for cryptocurrency mining [1]
- Other motivations for deploying malware include espionage, activism, and fame among cybercriminal communities
- Attackers often target endpoints (devices connected to the network) to disrupt operations or use sensitive information for extortion [1]

Seven Major Malware Categories

Virus

Replicates by modifying other programs

Worm

Self-replicates and spreads across networks

Trojan

Disguised malware appearing legitimate

Ransomware

Encrypts files and demands payment

- Virus attaches to clean files and spreads throughout systems, infecting files with code that damages core functionality [1]
- Worms replicate themselves and spread to other computers without human action, exploiting network vulnerabilities to propagate rapidly
- Trojan horses masquerade as legitimate software while secretly providing backdoor access for data theft or system control [1]
- Ransomware encrypts data and demands ransom payment for the decryption key, locking users out of their systems [1]

More Malware Types

- Spyware operates in the background collecting keystrokes, browsing habits, passwords, and personal data without user knowledge
- Adware bombards users with pop-ups and redirects while tracking online behavior to build advertising profiles
- Browser hijackers change your homepage, search engine, and redirect web traffic to generate advertising revenue for attackers
- All three types compromise privacy and can serve as entry points for more dangerous malware infections

Spyware

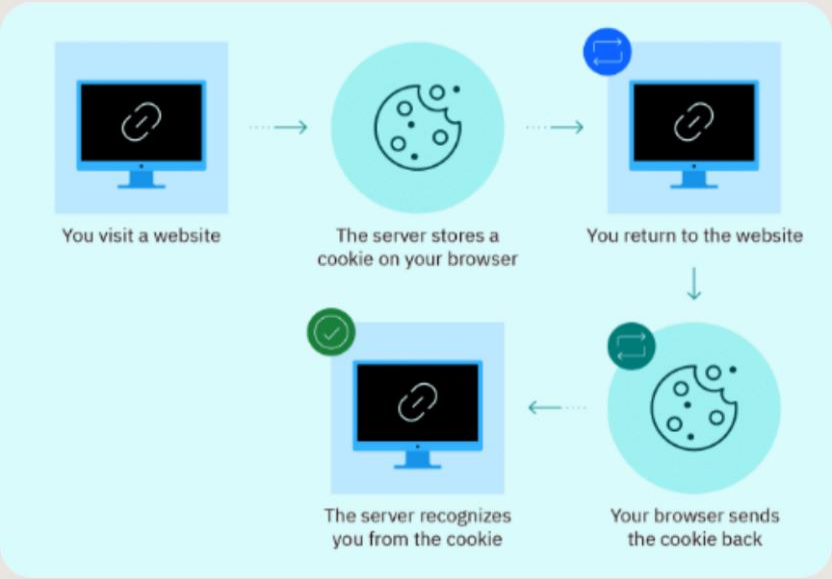
Secretly monitors and collects user information

Adware

Displays unwanted advertisements and tracks behavior

Browser Hijacker

Modifies browser settings without consent



Antimalware Software Protection

- Antimalware software detects, quarantines, and destroys malware through continuous scanning [1]
- Well-known programs include Malwarebytes, McAfee Antivirus, and Windows Defender [1]
- Detection works by scanning files for malware signatures, which are attribute patterns from known malware [1]
- After identifying malware, software deletes the file, quarantines it, or alerts the user [1]

Part 4: Social Engineering Attacks

What is Social Engineering?

Deception

Tricks individuals into divulging information

Access

Opens doors to sensitive systems

- Social engineering uses deception to trick individuals into divulging confidential or personal information that cybercriminals could use for fraud [1]
- After persuading someone to perform a particular action, attackers can access sensitive systems, steal assets, or perform more complex attacks [1]
- These attacks succeed because they exploit trust, authority, fear, and urgency rather than technical weaknesses
- Social engineering is often the first step in multi-stage attacks that combine human manipulation with technical exploits

Attack

Send fake message from trusted source

Deception

Create urgency with fake security alert

Capture

Victim clicks malicious link or enters credentials

Exploit

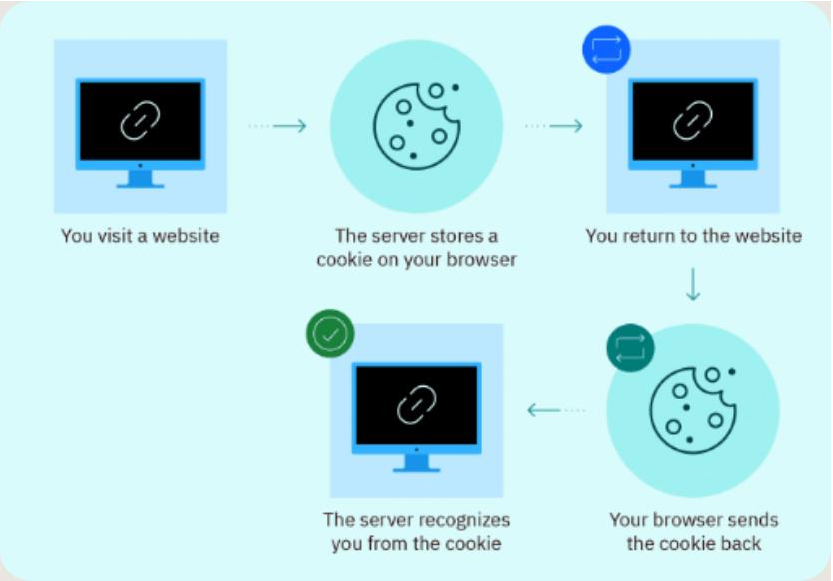
Attacker gains access to accounts and data

Phishing Attacks Explained

- Phishing is the practice of sending messages that trick users into providing sensitive information through fake alerts and malicious links [1]
- Example phishing attack: attacker sends fake data breach alert from your bank urging you to click password reset link that records your credentials [1]
- Organizations typically issue policies and training to help employees identify and report phishing attempts to reduce successful attacks
- Browser extensions like Malwarebytes Browser Guard block known malicious websites by default, lowering chances of visiting them during phishing attempts [1]

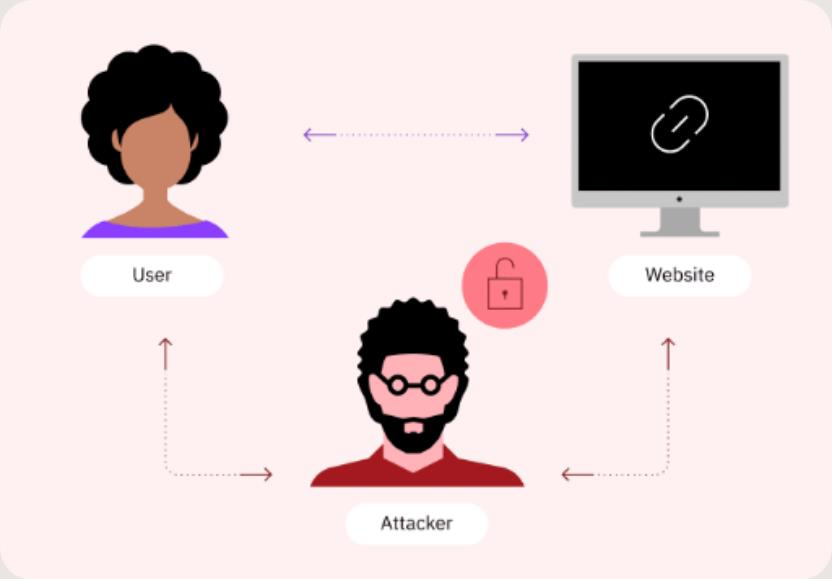
Cookies and Tracking

- A cookie is a data file containing information about your interaction with a website including username, password, login state, and preferences [1]
- When you return to a website, your browser sends the cookie back to the server which can personalize your experience and keep you logged in [1]
- Tracking cookies track your browsing activity across websites without consent, revealing which sites you've visited to third-party advertisers [1]
- Cookie hijacking occurs when attackers access first-party cookies containing sensitive information like login credentials to impersonate you on websites [1]



Part 5: Network-Based Attacks

Eavesdropping and Man-in-the-Middle



Unsecured networks allow attackers to intercept your communications

- Eavesdropping is when an attacker secretly accesses communications across an unencrypted network to steal sensitive data [1]
- Man-in-the-middle attacks involve attackers inserting themselves between two devices to capture and modify communications [1]
- Example scenario: attacker sets up free WiFi hotspot in public allowing them to view communications and redirect to fake login screens [1]
- You are especially vulnerable on unsecure or open networks that are free, unencrypted networks without password requirements [1]

Session Hijacking

- Session hijacking occurs when attackers take control of your active session in real time, such as transferring money while you're logged in [1]
- Attackers can steal session tokens through network eavesdropping, cross-site scripting, or malware on your device
- Once in control, attackers can perform any action you're authorized to do without needing your password
- Always log out when finished, especially on public or shared computers, and monitor account activity for unauthorized actions

Login

User authenticates to secure website

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Session

Browser maintains active authenticated session

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Hijack

Attacker steals session token or cookie

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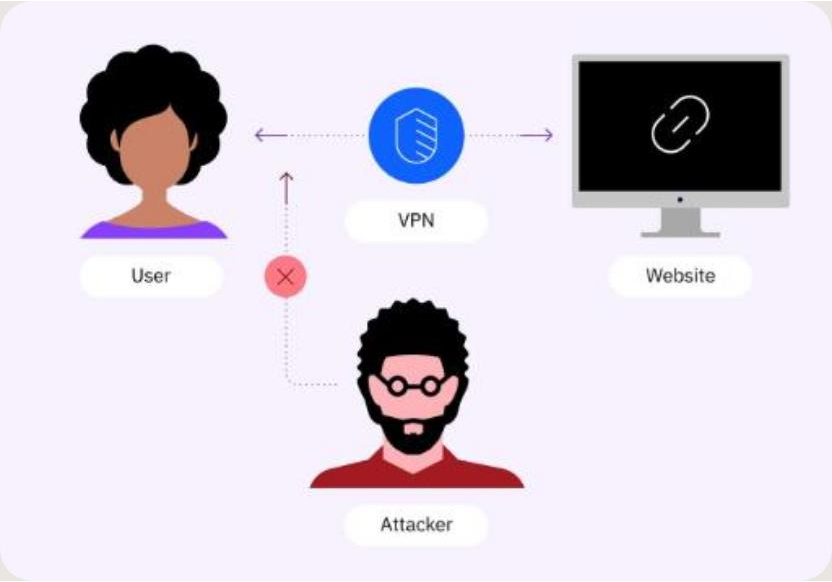
Control

Attacker performs actions as authenticated user



VPN Protection

- A VPN is a virtual network that encrypts communication between sites, protecting all data sent from your device including IP address [1]
- Your VPN encrypts all data from your device, then the recipient's VPN server decrypts it before accessing websites [1]
- Browse the web anonymously, preventing attackers, ISPs, and vendors from reading or decrypting your data [1]
- Choose a specific region for the VPN server to view region-specific content and bypass geographic restrictions [1]

Malwarebytes Protection Suite

- Malwarebytes provides antimalware scanning to detect and delete malicious files from your computer by identifying known malware signatures [1]
- AdwCleaner is a specialized tool within Malwarebytes that detects and removes adware and spyware that other scanners might miss [1]
- Privacy VPN conceals your device's online data through encryption, preventing attackers and ISPs from tracking your browsing activity [1]
- Browser Guard extension blocks online security threats and known malicious websites, providing real-time protection during web browsing [1]

Core

Antimalware scans and malicious file deletion

AdwCleaner

Specialized adware and spyware removal

Privacy VPN

Encryption for online data protection

Browser Guard

Real-time threat and tracker blocking

Part 6: Your Cybersecurity Career Path

The Career Opportunity

5.5M

Cybersecurity professionals working worldwide [1]

4.7M

Currently work in cybersecurity globally [2]

4M

Cybersecurity positions unfilled globally [1]

#7

Ranking among America's 100 best jobs [1]

Growth rate *is much faster than average for all occupations, creating* ***tremendous opportunities*** *for new professionals*

Industries Seeking Cybersecurity Talent

Finance & Healthcare

Banking, pharmaceuticals, insurance require security expertise

Technology & Retail

Computer tech, e-commerce, gaming drive innovation

Government & Infrastructure

Public sector, energy, utilities protect critical systems

Specialized Industries

Education, manufacturing, aerospace, hospitality all need protection

- Nearly all organizations need cybersecurity professionals regardless of their business focus or industry sector [1]
- Financial, government, education, and retail sectors are prominent players due to size and data sensitivity
- Emerging sectors like agriculture, construction, and automotive are increasing cybersecurity investments
- Cross-industry demand provides flexibility to align your career with personal interests

Cybersecurity Analyst Role

Entry to Advanced

Position levels available [1]

- Primary responsibility is to protect systems, networks, and data from cyberthreats through continuous vigilance [1]
- Monitor systems for unusual activities or security breaches, acting quickly to resolve issues
- Create defensive strategies and develop incident response protocols to keep data safe
- Role scope varies by organization, available at entry-level, mid-level, or advanced positions

Engineering and Architecture Roles

Mid to Advanced

Typical position levels [1]

- Cybersecurity Engineer designs and builds security architectures to protect sensitive information through monitoring and testing [1]
- Engineers conduct routine testing and simulations to identify and mitigate vulnerabilities
- Network Security Architect designs secure networks that withstand cyberthreats, like an architect designing a building [1]
- Architects develop network security measures, regularly updating them as threats emerge and helping recover networks after breaches

Development and Testing Roles

Developers and testers ensure software security throughout the development lifecycle

- Security Software Developer designs and develops secure software integrating security measures into all stages of the software development lifecycle [1]
- Developers update and patch existing software to reinforce defenses as new threats emerge
- Penetration Tester (Pen Tester) are ethical hackers who simulate real hacking techniques to find vulnerabilities in digital systems [1]
- Pen testers confirm vulnerabilities identified in scans and uncover weaknesses missed by automated assessments

Specialized Analysis Roles

- Malware Analyst investigates suspicious files or programs that might be malware, studying them closely to determine how they work and how to stop them [1]
- This role requires deep knowledge of computers and programming languages plus an aptitude for problem solving and staying informed on latest malware threats
- Digital Forensics Investigator are digital detectives who retrieve, examine, and analyze digital evidence to investigate cybercrimes after they are committed [1]
- Use forensic tools and deep understanding of attack frameworks to examine digital data, recover lost or damaged files, and trace digital footprints back to cybercriminals

Malware Analyst

Investigates malicious files to understand and stop them (Mid to Advanced)

Digital Forensics

Retrieves and analyzes digital evidence from cybercrimes (Entry to Advanced)

Application Security

Works with developers on security throughout development lifecycle

Entry Level (0-2 years)

Install security tools, manage services, report events

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Mid Level (3-5 years)

Maintain data integrity, review assessments, recommend improvements

↓

Management (5+ years)

Lead teams, oversee strategy, architecture, incident response

Career Progression Paths

- Entry-level technicians install security tools, manage services, report events, and help users identify threats [2]
- Mid-level specialists maintain data integrity, review risk assessments, and recommend security improvements [2]
- Directors lead security teams and oversee strategy, architecture, administration, and incident response [2]
- Specialization options include administration, engineering, forensics, penetration testing, and security analysis

Part 7: Building Your Skill Set

Baseline Technical Skills

Information Security

Protecting confidentiality, integrity, and availability of data

Threat Analysis

Identifying, analyzing, and evaluating potential threats

Cryptography

Securing information and verifying user identities

Risk Management

Assessing organizational risks and vulnerabilities

- Information Security and Assurance involves protecting and ensuring the confidentiality, integrity, and availability of data from unauthorized access, use, or modification [1]
- Security assurance verifies that security controls like firewalls, intrusion detection systems, and encryption protocols effectively protect networks and devices from threats [1]
- Cryptography secures information by converting it into code that only authorized people or systems can decode, protecting sensitive data during transmission [1]
- Risk management and assessment involves identifying, assessing, and limiting risks that organizations face by analyzing potential vulnerabilities and their impacts [1]

Network and Authentication Skills

Network Security:

- Design and implement secure network architectures
- Configure firewalls, intrusion detection systems, and secure protocols
- Monitor network traffic for suspicious activity and anomalies

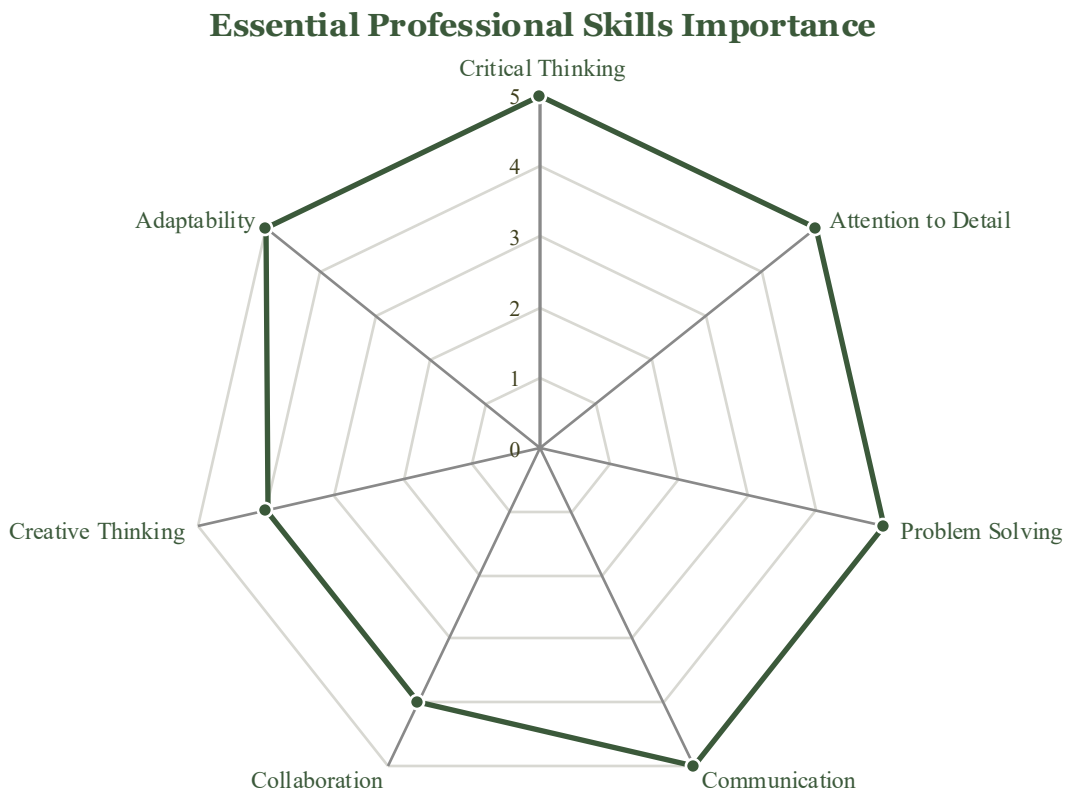
Authentication & Access:

- Implement multi-factor authentication systems
- Manage identity and access management (IAM) solutions
- Ensure proper authentication verifies user identities securely

Workplace Professional Skills

Technical skills alone aren't enough—soft skills are equally critical for success

- Critical thinking allows you to analyze complex cybersecurity challenges, identify potential threats, and make informed decisions about addressing them effectively [1]
- Attention to detail helps you carefully observe security measures, processes, and data since a subtle clue in a log file might be key to identifying a breach [1]
- Communication skills enable you to convey complex technical information clearly to both technical and non-technical audiences through reports and presentations [1]
- Adaptability is essential as technology evolves at an incredible rate, requiring you to learn emerging technologies and implement new processes impacting cybersecurity [1]



Source: Your Future in Cybersecurity Careers Threats and Prot...

Specialized Skills and Tools

Malware Analysis

Investigate and neutralize malicious code

Cloud Security

Protect cloud infrastructure and services

Python

Essential programming for automation

Penetration Testing

Ethical hacking to find vulnerabilities

- Specialized skills include malware analysis, cloud security, Linux/Windows security, Python, DevOps, and pen testing [1]
- Cloud security expertise is increasingly critical as organizations migrate infrastructure and data to cloud platforms
- Python programming enables automation of security tasks, data analysis, and development of custom security tools
- Penetration testing skills allow you to think like an attacker to identify vulnerabilities before malicious hackers exploit them

Professional Certifications

- CompTIA offers respected certifications including Security+, Cybersecurity Analyst, PenTest+, and Advanced Security Practitioner [1]
- ISC2 provides professional certifications such as Certified in Cybersecurity, CISSP, and CCSP [1]
- EC-Council credentials include Certified Ethical Hacker, Certified Penetration Testing Professional, and Computer Hacking Forensic Investigator [1]
- Industry-specific knowledge is valuable, such as understanding HIPAA regulations for healthcare or compliance frameworks for financial services

CompTIA

Security+, CySA+, PenTest+, CASP+

ISC2

Certified in Cybersecurity, CISSP, CCSP

EC-Council

CEH, CPENT, CHFI

Getting Started in Cybersecurity

Expand Knowledge

Explore different cybersecurity roles and industries

↓

Continue Learning

Take online courses and build foundational skills

↓

Explore Opportunities

Research job postings to identify common qualifications

↓

Build Skills

Focus on baseline, workplace, and specialized skills over time

- Cybersecurity professionals come from diverse backgrounds, some from IT and some from different fields entirely [1]
- The key is to build relevant skills that give you basics to enter a role and grow from there [1]
- Free learning resources include Linux Foundation, IBM Security Learning Academy, NIST online content, and SANS Institute courses [1]
- Research job postings to identify common requests and qualifications for roles that appeal to you [1]

Learning Resources

Free Resources:

- Linux Foundation online courses and certifications
- IBM Security Learning Academy training modules
- NIST Cybersecurity Framework documentation
- SANS Institute cyber aces tutorials

Paid Learning Platforms:

- Coursera and edX professional certificates
- Udemy cybersecurity courses and bootcamps
- Pluralsight technology skill paths
- CompTIA and ISC2 official training materials

Part 8: Protecting Yourself Today

Strong Passwords

Use unique, complex passwords for every account

Software Updates

Install security patches and updates immediately

Safe Browsing

Verify website legitimacy before entering credentials

Phishing Awareness

Scrutinize unexpected emails and messages

Personal Security Best Practices

- Always practice safe browsing, detect phishing, update software regularly, and use strong unique passwords
- Use password managers to generate and store complex passwords for all accounts securely
- Enable automatic updates on all devices to ensure critical security patches arrive immediately
- Be skeptical of urgent requests for personal information, even from legitimate sources

Essential Security Tools

- Install reputable antimalware software like Malwarebytes, Windows Defender, or similar protection on all devices
- Use a VPN whenever connecting to public WiFi networks to encrypt your traffic and protect sensitive data
- Browser extensions like uBlock Origin and privacy-focused tools block trackers, ads, and known malicious websites
- Maintain regular backups of important data to external drives or cloud storage to recover from ransomware attacks

Foundation

Antimalware software on all devices

Network

VPN for public WiFi and privacy

Browser

Extensions that block trackers and malicious sites

Backup

Regular backups to protect against ransomware

Privacy Protection Strategies

Online Privacy:

- Review and adjust privacy settings on social media platforms
- Limit information shared publicly on profiles and posts
- Use privacy-focused search engines like DuckDuckGo
- Clear cookies and browsing history regularly

Data Protection:

- Understand what data organizations collect about you
- Exercise your rights under privacy regulations like GDPR
- Review terms of service before accepting
- Monitor accounts for unauthorized access or suspicious activity

Your Cybersecurity Journey

- Cybersecurity careers are fastest-growing with **32% projected growth by 2032** and **4 million unfilled positions globally** [1]
- Build baseline skills in **security, cryptography, threat analysis, and risk management** combined with communication and adaptability [1]
- Pursue certifications from **CompTIA, ISC2, or EC-Council** to accelerate learning and open advanced positions [1]
- Protect organizations by understanding malware, phishing, eavesdropping, and using antimalware software and VPNs [1]

Key Takeaways and Next Steps

3

Key Elements (People, Processes, Technology) protect systems effectively

7

Major malware categories to defend against [1]

10

GAPP privacy principles guide best practices [2]

35%

Cybersecurity career growth by 2031 [2]

*Understand that you play a **critical role** in cybersecurity at home, school, or workplace through your **choices***

Understanding Malware Threats

- Attackers design malware to remain undetected while using device resources to launch attacks, host illicit data, or access information [1]
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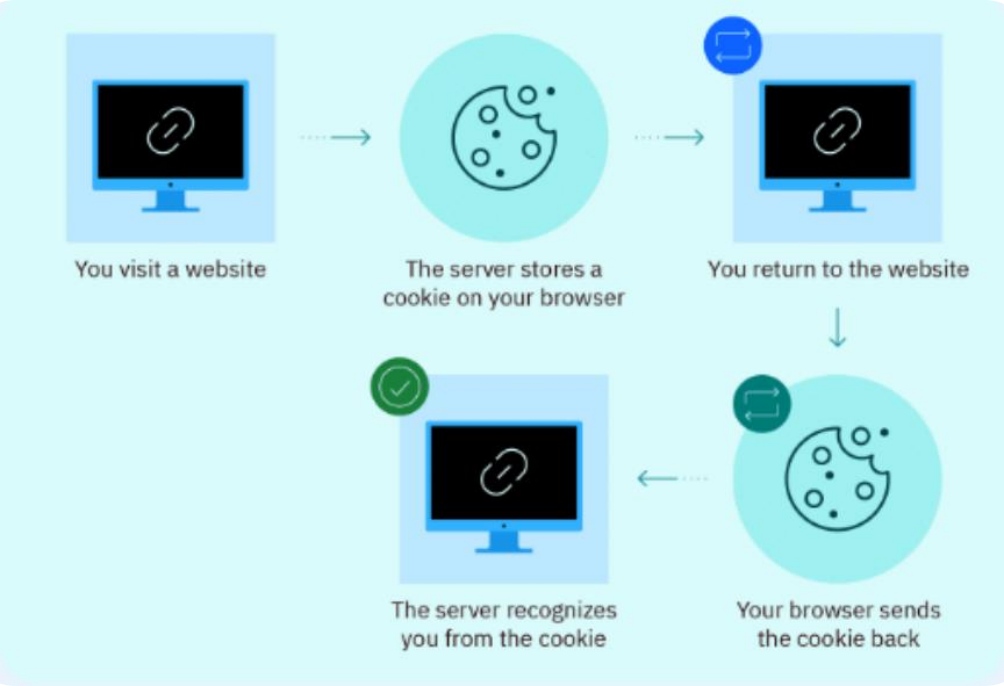
Antimalware Software Protection



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- Detection works by scanning files for malware signatures, which are attribute patterns from known malware [1]
- After identifying malware, software deletes the file, quarantines it, or alerts the user [1]
- EDR systems send logs and scan results to centralized servers for organizational analysis [1]

Social Engineering Attacks

- Social engineering uses deception to trick individuals into divulging confidential or personal information that cybercriminals could use for fraud [1]
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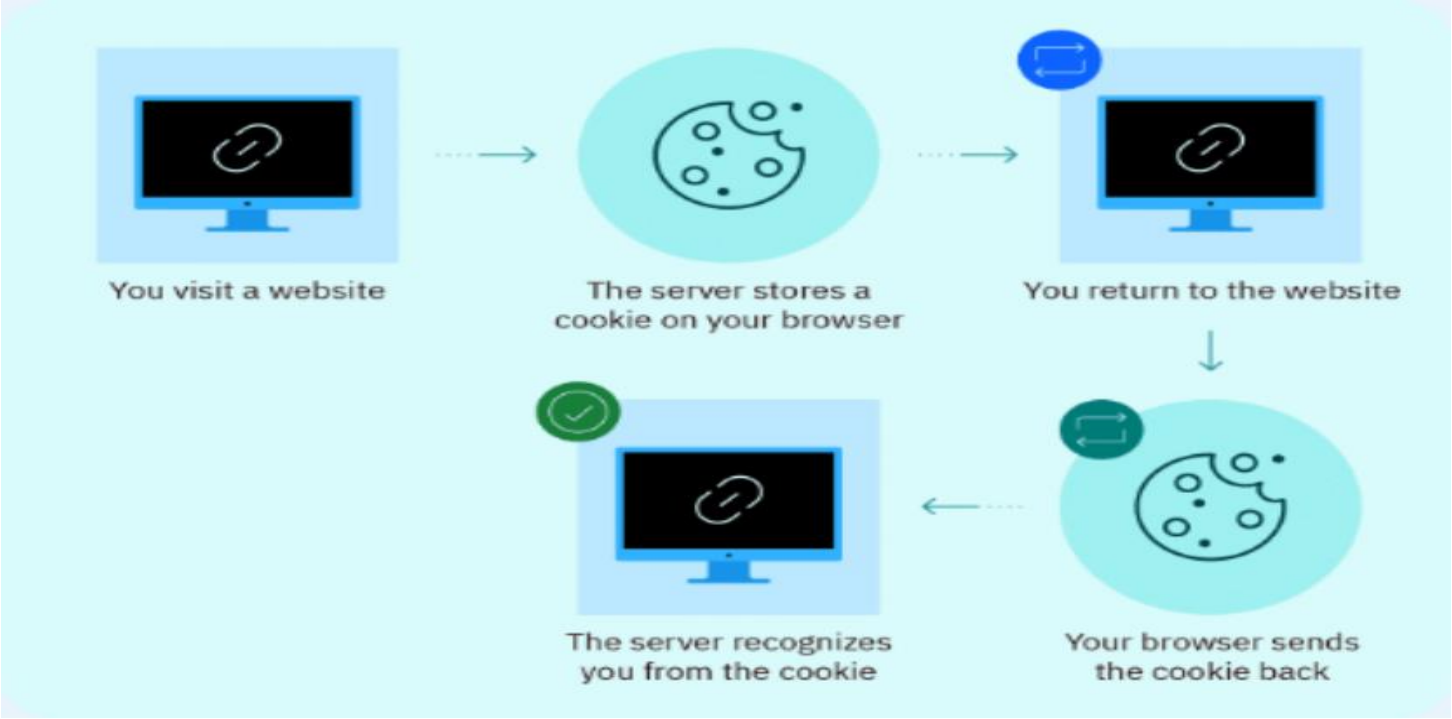
- Browser extensions like Malwarebytes Browser Guard block known malicious websites by default, lowering chances of visiting them during phishing attempts [1]

Understanding Cookies and Tracking

- A cookie is a data file containing information about your interaction with a website including username, password, login state, and preferences [1]
- When you return to a website, your browser sends the cookie back to the server which can personalize your experience and keep you logged in [1]
- Tracking cookies track your browsing activity across websites without consent, revealing which sites you've visited to third-party advertisers [1]
- Trackers can monitor your IP address and location, with companies using this information to display targeted ads that could be harmful if exposed [1]
- Cookie hijacking occurs when attackers access first-party cookies containing sensitive information like login credentials to impersonate you on websites [1]
- Today's browsers and extensions like Malwarebytes Browser Guard offer options for blocking trackers and protecting your privacy [1]



References

Here are trusted cybersecurity reference links you can use for learning and staying updated:

•Cybersecurity & Infrastructure Security Agency (CISA)
https://www.cisa.gov
•National Institute of Standards and Technology (NIST) – Cybersecurity Framework
https://www.nist.gov/cyberframework
•OWASP (Open Web Application Security Project)
https://owasp.org
•SANS Institute – Cybersecurity Training & Resources
https://www.sans.org
•IBM Cybersecurity Learning Hub
https://www.ibm.com/security
•Cisco Networking Academy – Cybersecurity Courses
https://www.netacad.com/cybersecurity
•Malwarebytes Security Blog
https://www.malwarebytes.com/blog
•Krebs on Security (Cybersecurity News & Analysis)
https://krebsonsecurity.com

Thank You